

FIML goodness-of-fit under non-normality: FMG/pEBA vs the MLR default

Question

The dominant default for incomplete continuous SEM is lavaan's `estimator = "MLR"`, `missing = "ml"` — the Mplus variant of the Yuan-Bentler mean-scaled FIML test. Under non-normality the test statistic's reference law (a weighted sum of χ_1^2) spreads its eigenvalues, so two questions are separable: does MLR's finite-sample trace approximation match exact saturated-space mean scaling, and does the Foldnes-Moss-Gronneberg eigenvalue family (scaled-shifted, EBA/pEBA, penalized-all, the full mixture) control the goodness-of-fit Type-I rate where MLR does not?

Short answer

Under **MCAR** (where normal-theory FIML stays consistent), yes. At severe Vale-Maurelli non-normality the MLR default rejects a true model **0.438** of the time at the .05 level. Exact UT mean scaling (`YB_exact`, the same p-values as the FMG **SB** row) rejects **0.166 (0.141)** with structured H1 information). The full FMG mixture rejects **0.051 (0.029)** structured), and scaled-shifted rejects **0.061 (0.036)** structured). The pattern holds across all non-normal families (5000 reps).

Under this study's **MAR** the same conclusion holds. The Savalei-Bentler mechanism here deletes indicators by coarse *sign* rules on two always-observed indicators, which is mild enough that FIML stays effectively unbiased: MAR rejection tracks MCAR for every method at a matched rate, and the well-calibrated tests stay at level under both. That is a property of *this weak* MAR, not of MAR in general. A stronger mechanism (steep continuous selection, or a genuinely nonlinear missing-given-observed mean) would bias FIML itself and leave a residual no reference-law correction can remove; this design does not probe that regime.

The model

One structural model is run throughout: a single common factor with six indicators $x_1, \dots, x_6$ in two groups ($n = 500$ and 350). The population is **scalar-invariant** – equal loadings and equal intercepts across groups – with a group-2 factor-variance and latent-mean shift, so the fitted models are correctly specified under the null. Each replicate is fit at two nested invariance levels, **configural** (loadings free across groups) and **metric** (loadings tied). We report two likelihood-ratio tests, and the **metric model is the null H_0 in both**; only the alternative differs:

- **goodness of fit** – metric vs the *saturated* model ($df = 23$): does the one-factor metric model fit the data at all? This is the ordinary test of exact fit (the “model chi-square”);
- **nested invariance** – *configural* vs metric ($df = 5$): the chi-square-difference test of the five equal-loading constraints.

Both are likelihood-ratio tests of one model nested in a larger one (metric in the saturated; metric in configural); the unqualified term “goodness of fit” means the first. Power is created by two local misspecifications, one per test: a residual covariance between x_5 and x_6 (0.13) that the one-factor model omits (powers the goodness-of-fit test), and a group-2 loading inflated $\times 1.35$ on one indicator (loading non-invariance; powers the nested test). Non-normality (the VM/IG/PL families) and Savalei-Bentler MCAR/MAR missingness act on the generated indicators; the factor model is otherwise unchanged.

What is tested

Each fit maximizes the observed-data normal (FIML) log-likelihood over the saturated moment vector $\eta = (\mu, \text{vech } \Sigma)$ with score $h_i(\eta)$. The saturated estimator $\hat{\eta}$ has the **sandwich covariance** (the “ Γ matrix”)

$$\Gamma_{\text{mis}} = H^{-1} J H^{-1}, \quad H = -\mathbb{E}[\partial h_i / \partial \eta'], \quad J = \mathbb{E}[h_i h_i'],$$

the missing-data analogue of Browne’s ADF moment covariance. If the working normal model is correct, information equality gives $J = H$, so $\Gamma_{\text{mis}} = H^{-1}$; under non-normality J carries the third- and fourth-moment information and $\Gamma_{\text{mis}} \neq H^{-1}$.

Take the goodness-of-fit test first. The metric SEM implies $\eta(\theta)$ with Jacobian $\Delta = \partial \eta(\theta) / \partial \theta'$. For a curvature V the residual operator is $U(V) = V - V \Delta (\Delta' V \Delta)^{-1} \Delta' V$, and the FIML likelihood-ratio against the *saturated* fit, $T = 2(\hat{\ell}_{\text{sat}} - \hat{\ell}_{\text{metric}})$, has the limit

$$T \Rightarrow \sum_{j=1}^{\text{df}} \lambda_j \chi_{1j}^2, \quad \lambda_j = \text{eig}_j\{U(V) \Gamma_{\text{mis}}\}.$$

When the normal model is correct $\Gamma_{\text{mis}} = V^{-1}$, $U(V) \Gamma_{\text{mis}}$ is a rank-df projector, every $\lambda_j = 1$, and $T \Rightarrow \chi_{\text{df}}^2$ – the only case in which the unscaled LRT is pivotal. The robust tests differ only in how they reduce the spectrum $\{\lambda_j\}$:

- **MLR / SB** – mean scaling, $c = \frac{1}{\text{df}} \sum_j \lambda_j = \frac{1}{\text{df}} \text{tr}\{U(V)\Gamma_{\text{mis}}\}$, refer T/c to χ^2_{df} ;
- **SS** – match the mean *and* the variance (scaled-and-shifted);
- **FMG** (EBA/pEBA, all) – keep the whole spectrum (Imhof / exact mixture).

A mean-only c is exact only when the λ_j are equal; non-normality disperses them, so MLR must over-reject.

The nested configural-vs-metric test ($\text{df} = 5$) is the same construction applied to the difference of the two models' residual operators (Satorra 2000); the spectrum reduction above carries over verbatim, and its results are in the **nested** rows of **results/summary_rejection.csv**.

Structured (V_S) vs unstructured (V_{US}). V is the curvature $U(V)$ is built from, and the two test variants differ only in where it is evaluated:

$$V_{US} = H(\hat{\eta}) \quad (\text{saturated fit}), \quad V_S = H(\eta(\hat{\theta})) \quad (\text{model-implied moments}).$$

The default columns use V_{US} ; the **_structured** columns use V_S . They share a probability limit under the correct MCAR null – so here they agree up to finite-sample terms – and separate only under misspecification or strong non-normal MAR. Full derivations: [docs/research/notes/non_normal_fiml_mlr_information](#).

Evidence

In the table below, columns are the data families: normal (the control), then Vale-Maurelli (VM), inverse-Gaussian (IG), and power-law (PL) generators at moderate (1; skew 2, kurtosis 7) and severe (2; skew 3, kurtosis 21) non-normality. Rows are the tests; **(str)** marks the structured- H_1 variant.

Table 1: GOF Type-I error (%), MCAR 30% (nominal 5%). Last column: mean over the six non-normal families. SB = YB; (str) = structured H1.

Method	norm	vm1	vm2	ig1	ig2	pl1	pl2	NN mean
naive	5.6	83.4	99.2	27.2	52.9	88.3	99.9	75.2
MLR	6.1	20.0	43.8	14.5	28.1	20.5	35.2	27.0
YB	5.9	10.8	16.6	7.6	9.1	10.6	16.9	11.9
YB (str)	6.4	9.7	14.1	7.5	11.5	8.7	14.7	11.0
SS	5.6	6.1	6.1	3.7	2.7	6.1	8.9	5.6
SS (str)	5.3	3.9	3.6	3.3	3.9	3.4	5.1	3.9
SF	5.5	5.5	5.2	3.4	2.4	5.6	8.2	5.0
EBA4	5.6	6.6	7.4	4.1	3.6	6.4	9.2	6.2
pEBA4	5.9	9.4	13.6	6.5	7.0	9.3	14.5	10.1
pEBA4 (str)	6.0	7.9	10.6	6.1	9.1	6.9	11.6	8.7
pall	5.9	9.1	12.0	6.1	5.8	9.0	13.9	9.3
pOLS	5.9	9.4	13.8	6.5	7.1	9.3	14.4	10.1
all	5.5	5.5	5.1	3.3	2.3	5.5	8.1	5.0
all (str)	5.2	3.3	2.9	3.0	3.4	3.0	4.5	3.3

Under MAR the same table tells the same story: the weak sign-based mechanism leaves FIML nearly unbiased, so the calibrated tests stay near 5%.

Table 2: GOF Type-I error (%), MAR 30% (nominal 5%). Last column: mean over the six non-normal families. SB = YB; (str) = structured H1.

Method	norm	vm1	vm2	ig1	ig2	pl1	pl2	NN mean
naive	5.2	85.2	99.2	25.0	46.8	88.5	100.0	74.1
MLR	6.1	18.8	39.7	13.8	25.2	17.4	34.3	24.9
YB	6.1	8.9	14.5	8.3	10.8	8.6	17.5	11.4
YB (str)	5.5	5.6	7.8	7.2	11.0	4.9	8.0	7.4
SS	5.5	4.9	4.7	4.3	3.4	4.5	7.8	4.9
SS (str)	4.4	1.9	1.3	2.9	3.6	1.1	1.6	2.1
SF	5.4	4.3	4.2	3.9	2.9	4.2	7.1	4.4
EBA4	5.5	5.4	6.1	4.7	4.4	4.7	8.2	5.6
pEBA4	5.9	7.7	11.7	7.0	8.5	7.4	14.3	9.4
pEBA4 (str)	5.2	4.4	5.2	5.7	8.2	3.7	5.4	5.5
pall	5.8	7.1	9.8	6.7	6.7	7.1	13.5	8.5
pOLS	5.9	7.7	11.8	7.0	8.6	7.4	14.3	9.5
all	5.4	4.3	4.0	3.8	2.7	4.2	7.0	4.4
all (str)	4.2	1.5	0.9	2.5	3.0	0.9	1.2	1.7

Table 3: GOF detection (%) at the severe cell (MCAR 30%, mean over the 6 non-normal families). Ordinary power tracks the Type-I inflation; size-adjusted power (scored at each test’s own empirical 5% cutoff) is essentially identical for every valid test, so MLR’s raw-power edge is purely over-rejection.

Method	Type-I	Power	Size-adjusted power
naive	75.2	91.9	18.8
MLR	27.0	51.9	17.4
YB	11.9	34.4	21.7
SS	5.6	22.1	21.8
SF	5.0	20.7	21.8
EBA4	6.2	23.5	21.8
pEBA4	10.1	31.1	21.8
pall	9.3	29.7	22.0
pOLS	10.1	31.2	21.8
all	5.0	20.5	21.8

Size-adjusted power scores each test at the cutoff that *would* give a 5% rejection rate of a true model in that cell, so an over-rejecting test cannot borrow detection from a broken Type-I rate.

On equal honest footing the valid tests have the same power; **naive** and **MLR** are no better (slightly worse), their apparent advantage being entirely the inflated size.

Nested test: configural vs metric (df = 5)

The same machinery applied to the measurement-invariance test (equal loadings) tells the same story. The competitors are the Satorra-2000 difference tests: **naive** (unscaled $\Delta\chi^2$), **SB** (scaled), **adjusted** (mean-and-variance), and **mixture** (exact). **naive** over-rejects badly, **SB** is mildly liberal, and **adjusted**/**mixture** hold near nominal.

Table 4: Nested Type-I error (%), MCAR 30% (nominal 5%). Last column: mean over the six non-normal families.

Method	norm	vm1	vm2	ig1	ig2	pl1	pl2	NN mean
naive	4.7	54.3	84.7	34.3	61.1	57.8	89.6	63.6
SB	4.7	7.5	11.7	5.7	7.3	7.7	10.2	8.3
adjusted	4.6	6.1	8.8	4.8	4.8	6.9	8.4	6.6
mixture	4.6	6.2	8.8	4.8	4.8	6.9	8.4	6.7

Table 5: Nested Type-I error (%), MAR 30% (nominal 5%). Last column: mean over the six non-normal families.

Method	norm	vm1	vm2	ig1	ig2	pl1	pl2	NN mean
naive	4.4	55.7	85.0	36.5	63.7	57.6	92.4	65.1
SB	4.5	7.4	10.8	6.3	7.2	7.2	12.0	8.5
adjusted	4.3	6.0	8.0	4.9	5.2	6.2	9.7	6.7
mixture	4.3	6.1	8.0	5.0	5.3	6.3	9.8	6.7

Table 6: Nested detection (%) at MCAR 30% (mean over the six non-normal families). As with goodness of fit, size-adjusted power is essentially equal across the valid tests; naive's raw-power edge is its over-rejection.

Method	Type-I	Power	Size-adjusted power
naive	63.6	95.3	42.3
SB	8.3	52.2	44.5
adjusted	6.6	48.4	44.2
mixture	6.7	48.6	44.2

Full per-cell rejection (all truths, mechanisms, rates, methods; goodness-of-fit and nested) is in `results/summary_rejection.csv`; per-replicate p-values, base statistics, and UGamma spectra are in `results/raw/` (the latter in `spectra.rds`) so any further test or size-adjustment is recomputable without refitting. The FIML-bias non-centrality diagnostic is in `results/noncentrality.csv` (it is a finite-sample non-normality artifact, equal for MCAR and MAR, not a MAR signal; see the note for why).

Caveats

- **One MAR, and a weak one.** The MAR cells use a mild sign-based Savalei-Bentler mechanism on always-observed indicators, so they behave like MCAR and do not probe the regime where MAR biases FIML itself. A stronger MAR is the obvious next run; MAR is not one mechanism.
- **Power is shown raw and size-adjusted.** The size-adjusted column is a diagnostic (its cutoff is read from the null simulation, unavailable in practice); it exists to compare the tests' intrinsic detection at equal size.
- The MLR column is magmaan's `fiml_robust_mlr`, matched to lavaan's `yuan.bentler.mplus` defaults (see oracle). `YB_exact` is the exact saturated-space trace scaling from the FIML UT spectrum; it duplicates `SB` by construction. lavaan offers no mean-and-variance test under FIML, so `SS` is a magmaan-only construction.

Reproduce

```
Rscript run_experiment.R --smoke # single hardest cell
Rscript run_experiment.R --lavaan-parity # full grid + oracle
Rscript run_experiment.R --cells truth=h0,dist=vm2,mech=MCAR,rate=0.3 --reps 1000
Rscript scripts/size_adjusted_power.R # size-adjusted power from existing raw/, no refit
```

Appendix: comparison to lavaan

The MLR over-rejection is genuine, not a magmaan computation error: on the worst cells lavaan's *own* MLR, fit on the same data every replicate, gives the identical rejection rate (agreeing per-replicate to $\sim 1e-5$), and the statistic carries ~ 1.7 - 2.3 x the variance of its reference chi-square, so a first-moment scaling cannot calibrate it.

Table 7: Oracle: magmaan vs lavaan MLR (estimator = MLR, missing = ml, defaults).

Quantity	Max magmaan - lavaan
<code>fiml_lrt_chisq</code>	6.74e-09

Quantity	Max magmaan - lavaan
mlr_chisq_scaled	7.35e-06
mlr_pvalue_scaled	4.81e-07

Table 8: MLR is not a magmaan artifact: fitting lavaan’s own MLR on the same data every replicate (not just rep 1) reproduces the rejection rate exactly. The variance-inflation factor (>1) is why a mean-scaling must over-reject.

dist	MLR reject (magmaan)	MLR reject (lavaan)	max $ \Delta p $	variance inflation
vm2	0.420	0.420	1.3e-05	2.08
pl2	0.393	0.393	2.2e-05	1.71
ig2	0.287	0.287	1.2e-05	2.33